

Master Project

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Multi-Modal Prediction of Major Depressive Disorder

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1 Introduction

1.1 Major Depressive Disorder

Major Depressive Disorder (MDD) is a mental health condition that significantly impacts individuals' daily functioning and overall quality of life. Characterized by persistent feelings of sadness, hopelessness, and a lack of interest or pleasure in activities, MDD affects millions of people worldwide. Traditional methods for diagnosing and monitoring depression normally involve clinical interviews and standardized questionnaires, which can be subject to biases and limitations. As a result, there is a gradually-increasing demand for innovative approaches that leverage technology to provide real-time, objective assessments of mental health [1].

1.2 Predictive Modeling on Smartwatch Data

The advent of wearable technology, particularly smartwatches, has opened up new avenues for continuous monitoring of physiological and behavioral indicators associated with mental health conditions, including MDD. These devices are equipped with a variety of sensors that can capture a range of data, such as heart rate variability (HRV), electrodermal activity (EDA), blood volume pressure (BVP), sleep patterns, and physical activity levels. By utilizing these diverse data streams, researchers aim to develop more accurate and timely methods for detecting depressive symptoms and tracking their progression over time.

1.3 Key Metrics

Blood Volume Pressure (BVP) and, by extension, Heart Rate Variability (HRV) are key physiological measures that reflect the autonomic nervous system's regulation of heart function. Studies have shown that lower HRV is associated with increased stress and emotional dysregulation, both of which are often observed in individuals with MDD [2]. By analyzing BVP, and by association, HRV patterns, researchers can gain insights into an individual's stress response and emotional state, making it a valuable tool for early detection of depressive episodes.

Electrodermal Activity (EDA), which measures the skin's conductance response, serves as an indicator of arousal and emotional activation. Changes in EDA can signal emotional responses to stimuli, including stress and anxiety, both of which frequently co-occur with depression. Monitoring EDA through wearable devices can provide real-time feedback on an individual's emotional state, offering potential insights into depressive symptoms that may not be captured through self-reports [3].

In addition to HRV and EDA, other data modalities collected by smartwatches contribute to a more holistic understanding of MDD. Sleep patterns, measured through heart rate and movement data, are closely linked to mood disorders, as individuals with MDD often experience disrupted sleep. By tracking sleep quality and duration, researchers can identify correlations between sleep disturbances and depressive symptoms.

Skin temperature too plays an important role, as does the core body temperature. Where metrics like HRV can be complemented by it. Successful HRV bio-feedback could increase finger temperature [4]

Furthermore, physical activity levels, which can be monitored via accelerometers in smartwatches, play a crucial role in mental health. Regular physical activity is known to have a positive effect on mood and can be an important component of depression management. Conversely, decreased activity levels can be a marker of depressive episodes. By examining physical activity trends in conjunction with other physiological data, researchers can enhance their understanding of the behavioral patterns associated with MDD.

The integration of these various data streams allows for a more comprehensive approach to detecting MDD. Machine learning techniques are increasingly being employed to analyze this complex data, enabling the identification of patterns and predictors of depressive symptoms. By training algorithms on multi-modal datasets, researchers aim to create predictive models that can alert individuals or healthcare providers to potential depressive episodes, facilitating timely interventions.

Despite the promising potential of wearable technology in mental health assessment, several challenges remain. Issues related to data privacy, the need for robust validation studies, and the generalizability of findings across diverse populations must be addressed to ensure that these tools are both effective and ethical. Additionally, the interpretability of machine learning models in the context of mental health remains a critical area for ongoing research.

2 Related work

Shah et al. outlines Major Depressive Disorder (MDD) with specific criteria for diagnosis. Key points include symptoms, exceptions, and specifiers to determine the nature of the episode, such as severity (mild, moderate, severe) and presence of psychotic features. The manual notes that MDD is a common disorder, with significant impacts on functioning and quality of life [1].

The rise of wearable technologies and the abundance of physiological data have opened up promising avenues for using machine learning (ML) to detect and monitor mental health disorders, such as Major Depressive Disorder (MDD). An effective machine learning pipeline for MDD detection must involve multiple components: data collection, preprocessing, feature extraction, model selection, and hyper-parameter optimization. This survey synthesizes research from various domains of hyper-parameter optimization, physiological data processing, and stress detection, providing a comprehensive overview of the methodologies and tools that can contribute to a robust and effective pipeline for MDD detection.

2.1 Wearable Devices and Physiological Data Collection

A crucial step in any machine learning pipeline aimed at detecting MDD is the collection of relevant physiological data. Campanella et al. present a dataset collected using the Empatica E4, a wearable device that measures physiological signals such as photoplethysmography (PPG) and electrodermal activity (EDA)

[5]. These signals are highly relevant for detecting mental health conditions, as changes in heart rate and skin conductance can correlate with stress and emotional states. The dataset collected by Campanella et al. provides a comprehensive repository of physiological data that could be used to model stress responses, a key factor in understanding depression.

The pipeline proposed by Campanella et al. can serve as a model for developing similar approaches in MDD detection. Specifically, the use of a multi-modal approach—where PPG and EDA signals are combined—can provide a more holistic understanding of physiological responses associated with depressive episodes. Furthermore, this research introduces a protocol to simulate real-world stressors in a lab setting, allowing for controlled yet realistic data collection. This protocol could be adapted for MDD studies by focusing on scenarios likely to trigger depressive symptoms or stress responses in individuals with MDD.

2.2 Data Preprocessing and Feature Extraction

Once data is collected, effective preprocessing and feature extraction are critical for ensuring that the machine learning models can learn meaningful patterns. In the paper by Shekhar et al. (2022), the authors discuss various hyper-parameter optimization techniques and highlight the importance of efficient preprocessing and filtering in machine learning pipelines [6]. This is particularly relevant when dealing with noisy physiological signals from wearable devices, as noise and artifacts can significantly impact model performance.

In the context of MDD detection, preprocessing steps might include filtering to remove motion artifacts from PPG signals or applying smoothing algorithms to EDA data. Both Campanella et al. and Shekhar et al. emphasize the need for robust preprocessing frameworks to handle large amounts of noisy data. For example, the filtering techniques employed in the preprocessing of the Empatica E4 dataset can be integrated into an MDD detection pipeline to ensure high-quality input data [5] [6].

2.3 Model Selection and Hyperparameter Optimization

Selecting the right model and tuning its hyperparameters are central to the success of a machine learning pipeline. Shekhar et al. (2022) compare various hyper-parameter optimization tools, such as Optuna, HyperOpt, Optunity, and SMAC, across different machine learning models [6]. Hyperparameter tuning is especially important in MDD detection, as models such as random forests, gradient boosting machines, and neural networks can be highly sensitive to their hyper-parameters.

Bayesian optimization methods, as implemented in tools like Optuna and HyperOpt, provide an efficient way to navigate the hyper-parameter space and select optimal configurations. Shekhar et al. found that Optuna performed particularly well in balancing model performance and runtime, making it a suitable candidate for optimizing machine learning models in MDD detection pipelines. By dynamically adjusting hyper-parameters, tools like Optuna could help improve the generalization of models, especially when working with high-dimensional physiological data.

In addition to model selection, evolutionary algorithms (e.g., Optunity’s particle swarm optimization) offer an alternative approach to hyper-parameter optimization for neural network models, which could be useful when designing deep learning models for MDD detection. Given that the physiological data collected from wearables is often high-dimensional, the ability to quickly and efficiently explore the hyper-parameter space is crucial [6].

2.4 Stress Detection as a Proxy for Depression Symptoms

Detecting stress is an important component of understanding MDD, as stress often precedes or exacerbates depressive episodes. The work by Campanella et al. (2024) on stress detection using wearable sensors provides a framework for capturing stress-induced physiological changes that could be adapted for MDD detection [5]. Specifically, the methods used to monitor physiological responses to stress (e.g., changes in EDA and PPG signals) could be repurposed to detect patterns indicative of depression.

By leveraging stress detection methods, a machine learning pipeline for MDD could infer the mental state of individuals based on real-time data, enabling early intervention. This approach is supported by Shekhar et al. (2022), who discuss the application of ML algorithms in real-world scenarios where stress monitoring and emotional state detection are crucial [6]. In the context of MDD, such a system could monitor changes in stress levels over time and correlate these changes with self-reported depression symptoms or clinical assessments.

2.5 Reproducibility and Open Science

Reproducibility is a key concern in any machine learning pipeline, particularly in healthcare applications. The work on RAPIDS by Vega et al. highlights the importance of reproducible data pipelines for processing large datasets from mobile and wearable devices [7]. A pipeline for MDD detection must be transparent and easily reproducible, ensuring that results can be validated and replicated across different studies and clinical settings.

Furthermore, open-source platforms like RAPIDS offer modular and scalable workflows that can be adapted for MDD detection. For example, RAPIDS’ feature extraction and visualization tools could be used to develop a robust framework for visualizing physiological changes over time in patients with MDD [7]. By leveraging such open-source tools, researchers can build on existing frameworks to create scalable and reproducible pipelines for MDD detection.

2.6 Challenges and Future Directions

Despite the promising developments in hyper-parameter optimization, stress detection, and wearable data collection, several challenges remain in building an effective MDD detection pipeline. One key challenge is managing the variability in physiological responses across individuals. As noted in Campanella et al. (2024), different subjects may exhibit different stress or emotional responses, and the same is likely true for depression-related physiological changes [5]. Therefore,

the models must be flexible enough to account for individual differences in baseline physiological states.

Another challenge lies in integrating multi-modal data (e.g., combining wearable sensor data with clinical assessments or self-reported measures) to improve the accuracy and reliability of the model. Future research should explore the use of ensemble learning techniques or hybrid models that can process different data types in parallel.

3 Datasets

3.1 Pilot Data from the “Klinik für Psychiatrie und Psychotherapie” of the University Medical Center Freiburg

The study involved presenting participants with a series of stimuli across different tasks, designed to evoke a range of emotional and cognitive responses. Data was collected through Empatica sensors, video, and audio recordings. The tasks were structured as follows:

Standardized Image Presentation

A set of 52 images from the OASIS dataset, commonly used for emotion research, were displayed. Each image was shown for 5 seconds with a 1-second pause in between.

Abstract Image Presentation Participants were shown 11 abstract images featuring human facial characteristics. The duration between each image varied to examine participant reactions to irregular timing.

Storytelling For this task, participants were given a single image to view for 60 seconds. During this time, they were asked to describe the story and meaning they inferred from the image.

PANAS Questionnaire A self-reported measure, the Positive and Negative Affect Schedule (PANAS), was administered to assess the participant’s emotional state.

Stroop Test To elicit stress, participants performed the Stroop Test, in which they were asked to vocalize the color of words that appeared on the screen. A total of 50 stimuli were presented.

Video Presentation Eight videos from EmoStim (a dataset known for emotional stimuli) and one Fail compilation video were shown. The videos had varying lengths to capture dynamic emotional responses.

3.1.1 Data Collection

The key modalities recorded are:

- **Media:**
 - Video recorded with a GoPro Hero 12 at 120Hz frame rate and 1920x1080 resolution.
 - Audio sampled at 48,000Hz.
- **Sensor Data:**
 - Empatica device measurements:

- * Blood Volume Pulse (BVP) at 64Hz.
- * Accelerometer data at 64Hz.
- * Electrodermal Activity (EDA) at 4Hz.

- **Additional Information:**

- A timesheet (`experiment_timesheet.csv`) logs the experiment details.
- Data also includes synchronization tags marked by the user, indicating three events before the experiment.

3.2 WESAD

The dataset was collected from a group of participants under various conditions designed to elicit different emotional responses, making it a valuable resource for researchers looking to explore the interplay between physiological signals and psychological states.

The WESAD dataset comprises a similar set of modalities to the pilot study like the BVP, EDA, and accelerometer data. It also provides more modalities linked with the RespiBAN like ECG, and the EMG

Data collection for the WESAD dataset involved a series of controlled experiments where participants engaged in different activities, including resting, watching a movie, and completing a stress-inducing task. This design facilitates the exploration of how physiological signals vary across different contexts. Participants were equipped with wearable sensors to capture real-time data, leading to a rich dataset that encompasses both baseline and induced stress responses.

The dataset involves physiological data collected from 17 subjects (with two subjects excluded due to sensor malfunction) using two devices: a chest-worn device (RespiBAN) and a wrist-worn device (Empatica E4).

For each subject, the dataset is organized into folders that include:

- **Readme file (`SX_readme.txt`):** Contains subject-specific information and data quality notes.
- **Self-report questionnaires (`SX_quest.csv`):** Includes protocol schedules and responses to various psychological assessments such as PANAS, STAI, and SAM questionnaires, used to obtain ground truth for emotional states.
- **Raw sensor data from RespiBAN (`SX_respiban.txt`):** Contains physiological data from the chest-worn device.
- **Raw sensor data from Empatica E4 (`SX_E4_Data.zip`):** Contains wrist-worn device data.
- **Synchronized data (`SX.pkl`):** A pre-synchronized file containing all raw sensor data (both chest and wrist) and labels.

3.2.1 Data Collection

RespiBAN A chest-worn device that records data from various physiological sensors, including:

- Electrocardiogram (ECG)

- Electrodermal activity (EDA)
- Electromyography (EMG)
- Respiration (PZT sensor)
- Temperature
- Accelerometer (ACC)

All signals were sampled at 700 Hz.

Empatica E4 A wrist-worn device that provides data from:

- Accelerometer (ACC) sampled at 32 Hz
- Blood Volume Pulse (BVP) sampled at 64 Hz
- Electrodermal activity (EDA) sampled at 4 Hz
- Skin temperature (TEMP) sampled at 4 Hz

4 Variance Analysis

Intrapersonal Variance: Refers to the variability in physiological measurements within the same subject with time.

Interpersonal Variance: Refers to the variability across multiple subjects under the same experimental conditions.

Thus, intrapersonal variance helps in understanding how much a single subject's physiological responses fluctuate, while interpersonal variance reveals how different subjects respond differently under the same conditions.

In this section, we analyze the two datasets described in the previous section.

The Empatica data from the pilot dataset, for which variances are calculable include electrodermal activity (EDA), blood volume pressure (BVP), and temperature. For additional insights, we derive the heartrate variability (HRV) from the BVP data.

For the WESAD dataset, there are many more types of physiological data collected, divided into wrist and chest components, but owing to the data available from the pilot study, we limit the comparisons to the same metrics from the WESAD data as well.

One stark difference between the two is the number of subjects. The pilot study involves 3 participants, while for the WESAD data initially, 17 subjects participated in the study. However, due to sensor malfunction, data of two subjects (S1 and S12) had to be discarded [8].

4.1 Pilot Data

4.1.1 Intrapersonal Variances

Abstract Images

Subject	BVP	EDA	Temperature	HRV
1	0.001733	0.000003	0.000114	0.164035
2	0.004034	0.000005	0.001689	0.037137
3	0.002167	0.001000	0.007707	0.554219

Table 1: Intrapersonal Variances for Abstract Images

Panas

Subject	BVP	EDA	Temperature	HRV
1	0.000646	0.000003	0.003724	0.087044
2	0.004849	0.000149	0.018373	0.081379
3	0.005736	0.011347	0.011705	0.316673

Table 2: Intrapersonal Variances for Panas

Stroop Test

Subject	BVP	EDA	Temperature	HRV
1	0.000495	0.000002	0.001281	0.110962
2	0.011072	0.000113	0.007585	0.068214
3	0.002663	2.023173	0.001702	0.094333

Table 3: Intrapersonal Variances for Stroop Test

Storytelling

Subject	BVP	EDA	Temperature	HRV
1	0.001226	0.000002	0.001819	0.309150
2	0.004881	0.000015	0.003384	0.021305
3	0.002200	0.000198	0.003973	0.184549

Table 4: Intrapersonal Variances for Storytelling

Videos

Subject	BVP	EDA	Temperature	HRV
1	0.001303	0.000006	0.026232	0.115185
2	0.003206	0.000065	0.876223	0.165146
3	0.049474	0.160559	0.141826	0.733728

Table 5: Intrapersonal Variances for Videos

Standard Images

Subject	BVP	EDA	Temperature	HRV
1	0.000163	0.000002	0.024079	0.026224
2	0.003450	0.000028	0.010986	0.019426
3	0.001098	0.002471	0.009976	0.090680

Table 6: Intrapersonal Variances for Standard Images

4.1.2 Combined Intra-/Intrapersonal Variances

Experiment	Metric	Intra	Inter
Abstract Images	BVP	0.002644	1.494416e-06
	EDA	0.000336	3.309138e-07
	Temperature	0.003170	1.605954e-05
	HRV	0.251797	7.262015e-02
Panas	BVP	0.003744	0.000007
	EDA	0.003833	0.000042
	Temperature	0.011267	0.000054
	HRV	0.161699	0.018021
Stroop Test	BVP	0.004743	0.000031
	EDA	0.674429	1.364332
	Temperature	0.003523	0.000012
	HRV	0.091170	0.000464
Storytelling	BVP	0.002769	3.582669e-06
	EDA	0.000072	1.202882e-08
	Temperature	0.003059	1.239236e-06
	HRV	0.171668	2.083824e-02
Videos	BVP	0.017994	0.000744
	EDA	0.053543	0.008589
	Temperature	0.348093	0.212531
	HRV	0.338020	0.118063
Standard Images	BVP	0.001570	0.000003
	EDA	0.000834	0.000002
	Temperature	0.015014	0.000062
	HRV	0.045443	0.001546

Table 7: Intrapersonal and Interpersonal Variances Grouped by Experiment

4.1.3 Analysis**Abstract Images:**

HRV shows the highest intrapersonal variance (0.251797) among the metrics, indicating significant variability within subjects during this experiment. This

suggests that heart rate variability (HRV) is highly sensitive to the abstract images presented.

BVP, EDA, and Temperature exhibit much lower intrapersonal variances, with BVP having an intrapersonal variance of 0.002644, indicating more consistent responses within individuals.

Interpersonal variance is also highest for HRV (0.07262015), which implies significant differences between subjects in how their HRV responds to abstract images.

Panas:

HRV again shows significant intrapersonal variance (0.161699), suggesting considerable fluctuations in heart rate responses within individuals when exposed to stimuli in this condition.

Temperature shows an intrapersonal variance of 0.011267, indicating some variability in body temperature within subjects.

The interpersonal variance is still most prominent in HRV (0.018021), though not as high as in the abstract images condition, suggesting some consistency in how different subjects' HRV responded.

Stroop Test :

EDA shows an extremely high intrapersonal variance (0.674429), indicating significant fluctuations within subjects' electrodermal activity during the Stroop test, a task known to induce stress and cognitive load. HRV shows lower intrapersonal variance (0.091170) compared to the previous conditions. Interpersonal variance in EDA is very high (1.364332), suggesting substantial differences between subjects in their stress responses during the Stroop test.

Storytelling:

HRV again leads with high intrapersonal variance (0.171668), indicating variability in how subjects' heart rate variability responds during storytelling. EDA shows very low intrapersonal and interpersonal variances, indicating consistent responses both within and between subjects. The interpersonal variance in HRV (0.02083824) suggests moderate differences between subjects in how their heart rate variability responds.

Videos:

Temperature shows the highest intrapersonal variance (0.348093) among all metrics, indicating significant fluctuations in body temperature within subjects while watching videos. HRV and EDA also exhibit relatively high intrapersonal variances (0.338020 and 0.053543, respectively), indicating variability in responses. The Interpersonal variance is highest in Temperature (0.212531), suggesting considerable differences in how subjects' body temperatures respond to the video stimuli.

Standard Images:

This condition shows generally low intrapersonal variances across all metrics, with HRV having the highest at 0.045443. This suggests that standard images evoke more consistent physiological responses within individuals. Interpersonal variances are also low, with HRV again showing the highest (0.001546), indicating relatively consistent responses across different subjects.

4.1.4 Visualization

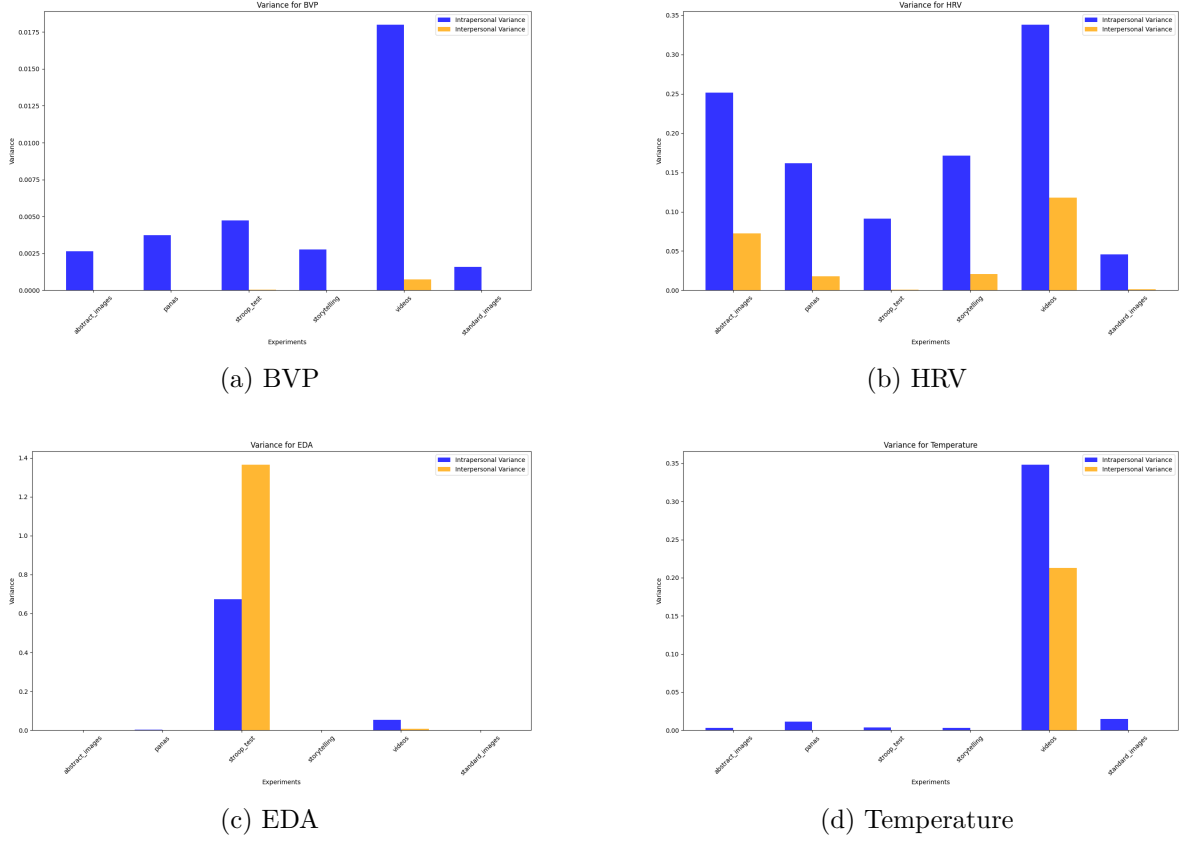


Figure 1: Intrapersonal and Interpersonal Variances Grouped by Experiment

4.2 WESAD Dataset

4.2.1 Analysis

Range and Variation:

BVP: Shows a relatively small range of values (from 0.000301 to 0.001661). This suggests that intra-subject variability in BVP is low, which is typical for physiological measurements where the signal is more stable within individuals.

HRV: The HRV values range from 0.070403 to 0.167851 across the subjects, indicating varying levels of heart rate variability among the individuals.

EDA: Shows moderate variability (from 0.016285 to 0.069296). This suggests some variability in EDA, which in turn suggests that it is a metric sensitive to emotional and physiological changes.

Temperature: Exhibits a range from 0.043823 to 0.147513. This range indicates variability in temperature measurements, due to different environmental conditions or body responses.

Patterns: Subjects S7, S8, S13, and S10 exhibit relatively higher HRV values (above 0.12). High HRV is generally associated with a well-balanced autonomic nervous system, reflecting good adaptability and resilience to stress.

Table 8: Intrapersonal Variances for Each Metric for the WESAD dataset

Subject	BVP	HRV	EDA	TEMP
S2	0.001661	0.083054	0.038760	0.081574
S3	0.000301	0.092467	0.037466	0.094117
S4	0.000762	0.070403	0.052117	0.092221
S5	0.001246	0.105681	0.045310	0.074219
S6	0.000626	0.080960	0.071279	0.079873
S7	0.000729	0.123586	0.055927	0.050710
S8	0.000762	0.119815	0.040599	0.090382
S9	0.000547	0.096164	0.029641	0.107554
S10	0.001109	0.167851	0.055965	0.062056
S11	0.001243	0.104607	0.040447	0.147513
S13	0.001427	0.130833	0.069296	0.110282
S14	0.001490	0.081998	0.000675	0.053648
S15	0.001012	0.120869	0.059420	0.043823
S16	0.000904	0.081243	0.056028	0.087088
S17	0.000873	0.075888	0.016285	0.068084

Table 9: Average Intrapersonal Variance and Interpersonal Variance for Each Metric

Metric	Intra	Inter
BVP	0.000979	1.353584×10^{-7}
HRV	0.102361	6.508775×10^{-4}
EDA	0.044614	3.373023×10^{-4}
TEMP	0.082876	6.651867×10^{-4}

These subjects may have a healthier cardiovascular profile or could be in a more relaxed or well-recovered state during the measurement.

Subject S14 shows a lower BVP value compared to others, possibly due to different physiological states or measurement settings. Consistent with physiological metrics, EDA and temperature show expected variability influenced by individual physiological and environmental factors.

S7 and S8 show high HRV, possibly indicating a good stress recovery capacity. S10 exhibits the highest HRV (0.167851) but moderate values in other metrics, suggesting that this individual might be in a very relaxed state or have a naturally high HRV.

S14 and S17 both show lower HRV values, with S14 also having the lowest EDA value, which might indicate low arousal or a flat autonomic response during the measurement.

Intrapersonal Variance: HRV exhibits the highest intrapersonal variance, highlighting its sensitivity to individual physiological states. BVP shows the lowest intrapersonal variance, indicating stability within individuals. **Interpersonal Variance:** HRV also has the highest interpersonal variance, reflecting significant differences between individuals. BVP, with the lowest interpersonal variance, suggests

minimal differences across subjects.

4.2.2 Visualization

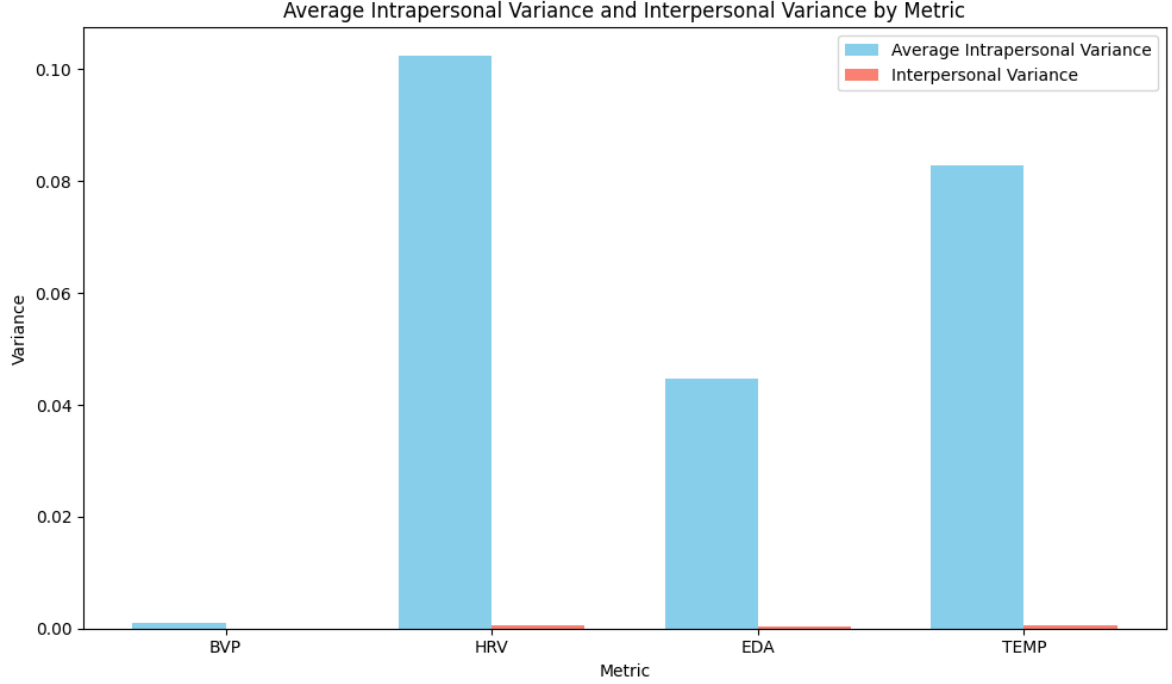


Figure 2: Plot 1

4.3 Summary

Thus, we can see a trend within both, the pilot and the WESAD datasets, wherein, the Intrapersonal variances are consistently much larger than the interpersonal variances, suggesting a similarity, albeit with some exceptions, in physiological and psychological conditions within the subjects recorded.

5 Implementation

For modelling, we focus on the WESAD dataset. We first build a preprocessing pipeline to appropriately prepare the data. Given especially the

5.1 Preprocessing Pipeline

5.1.1 Extraction

For our data, we make use of the pickle files stored in SX.pkl, containing the synchronized data and labels. The labels are described as ground truth labels in the documentation for the dataset.

The raw data from the two devices were manually synchronized using a specific double-tap signal pattern performed by the subjects during data collection. This

synchronization ensures that the physiological signals from both the chest-worn RespiBAN device and the wrist-worn Empatica E4 device are accurately aligned in time.

The synchronized data for each subject is provided in the file `SX.pkl`. This file is structured as a dictionary containing the following keys:

- **'subject'**: The identifier for the subject, labeled as `SX`.
- **'signal'**: Contains all the raw data organized into two fields:
 - **'chest'**: This includes all physiological modalities recorded by the RespiBAN device, specifically:
 - * ACC (Accelerometer)
 - * ECG (Electrocardiogram)
 - * EDA (Electrodermal Activity)
 - * EMG (Electromyography)
 - * RESP (Respiration)
 - * TEMP (Temperature)
 - **'wrist'**: This includes all physiological modalities recorded by the Empatica E4 device, specifically:
 - * ACC (Accelerometer)
 - * BVP (Blood Volume Pulse)
 - * EDA (Electrodermal Activity)
 - * TEMP (Temperature)
- **'label'**: The identifier for the respective study protocol condition, sampled at 700 Hz. The available labels are as follows:
 - 0 = Not defined / Transient
 - 1 = Baseline
 - 2 = Stress
 - 3 = Amusement
 - 4 = Meditation
 - 5/6/7 = Should be ignored in this dataset

For the purposes of this project, and in trying to follow the methodology from [8], we consider just the physiological data (i.e. ignoring the accelerometer data) and focus on building classification models for the states baseline, stress, and amusement.

Once the data is gathered, preprocessing is essential in making the raw physiological data usable for machine learning models.

5.1.2 Signal Interpolation

In the WESAD dataset, different sensors record data at different frequencies. For example, BVP is recorded at 64 Hz, while EDA and temperature are recorded at 4 Hz, and the labels are captured at 700 Hz. To align these signals, we need to interpolate them so that each signal has the same number of data points.

We use an interpolation function that resamples these signals to match the length of the labels. This ensures that each sensor's data is synchronized and corresponds directly to the same time intervals. - First, the original time vectors for the signals are calculated based on their sampling rates. - Then, interpolation is applied to the signals using the `scipy.interpolate.interp1d` function to create new signals that match the length of the labels.

This is important because without interpolation, the features from different sensors would not be aligned in time, leading to misrepresentation of the relationship between the physiological signals and the emotional states.

5.1.3 Feature and Label Extraction

After interpolation, the features are extracted from the three primary signals from the wrist: BVP, EDA, and temperature, and the five primary signals from the chest: ECG, EMG, EDA, temperature, and ResiBAN. These interpolated signals are combined into a single feature matrix, respective to the modality source, where:

- Each row represents a time point.
- Each column represents a different sensor signal.

This forms the input feature set for the model. For example, if we have 10,000 time points, we would end up with a feature matrix of size '10,000 x 3' (in the case of the three signals: BVP, EDA, Temp).

In the context of our study on predicting affective states, we carefully considered the features included in our predictive models. Ultimately, we chose to exclude accelerometer data (motion-based features) for the following reasons, in addition to matching the setup of Schmidt et al. in the physiological features only category:

Performance of Physiological Features: Research indicates that physiological features (such as heart rate variability, electrodermal activity, etc.) provide deeper insights into the emotional and psychological states of individuals compared to motion-based features. Specifically, Schmidt et al. noted from their results that classifiers trained on physiological data yield significantly higher classification scores than those based solely on accelerometer data. This suggests that physiological metrics more accurately reflect the underlying emotional responses, making them more valuable for our predictive objectives.

Avoiding Overfitting to Motion Patterns: Including only motion-based features could lead to models that primarily learn to distinguish between specific patterns of movement associated with the experimental conditions rather than genuinely understanding the subjects' affective states. Such models may perform well in training but fail to generalize to new data. By excluding accelerometer data, we mitigate the risk of our classifiers being biased toward motion patterns, thus improving the validity of our findings.

Focus on Relevant Insights: Our goal was to derive insights into the emotional states of subjects rather than merely understanding their physical activity levels. Physiological data are inherently linked to emotional arousal and stress, offering a more relevant basis for our predictions. In contrast, accelerometer data may not capture the nuanced emotional responses we aimed to analyze.

Simplified Model Complexity: Excluding accelerometer data allows us to simplify our models, reducing the dimensionality of the input data. This not

only enhances computational efficiency but also focuses the model on the most impactful features, facilitating clearer interpretations of results.

The decision to exclude accelerometer data was driven by a focus on maximizing predictive accuracy and ensuring that our models are robust and meaningful in assessing affective states. By prioritizing physiological features, we aim to provide more reliable insights into the emotional and psychological dimensions of our subjects.

Labels are collected during the experiment, indicating the emotional state at each time point (e.g., baseline, stress, amusement). The labels are already available in the dataset but need to be synchronized with the features, especially given the difference in frequencies as noted in the preprocessing section.

5.2 Modeling

5.2.1 Cross Validation

Leave-One-Subject-Out Cross-Validation (LOSOCV) is a specific case of cross-validation that involves partitioning a dataset into training and validation sets. For a dataset with N observations, LOSOCV trains the model on $N - 1$ samples, leaving one observation out for validation. This process is repeated N times, ensuring that each observation serves as the validation set exactly once. Given a dataset D with N instances, LOSOCV selects one instance D_i as the test set while using the remaining $N - 1$ instances for training. The model M is trained on $D - D_i$ and evaluated on D_i , resulting in N separate evaluations.

The performance metrics such as accuracies and F1-scores are calculated for each iteration, and the overall performance of the model is the mean of these N metrics, mathematically represented as:

$$\text{Performance} = \frac{1}{N} \sum_{i=1}^N L(D_i, M(D - D_i))$$

where $L(D_i, M(D - D_i))$ denotes the evaluation metric for instance D_i . LOSOCV typically has low bias since it utilizes nearly all available data for training in each iteration, which often leads to more stable models. However, it can exhibit high variance, especially when the dataset is small or contains outliers, making model performance sensitive to individual data points [9]. In our case, we have sufficiently large training data to prevent this.

5.3 Hyperparameter Optimization

In machine learning, the performance of models heavily depends on the selection of hyperparameters. Hyperparameter optimization (HPO) is an important step in enhancing model performance, as it involves fine-tuning the parameters that govern the learning process. Among various tools available for HPO, Optuna stands out due to its flexibility, efficiency, and user-friendly design.

Optuna is an open-source software framework specifically developed for hyperparameter optimization. It employs a sophisticated approach known as Bayesian

optimization, which iteratively explores the hyperparameter space to identify optimal configurations. This method allows Optuna to intelligently balance exploration and exploitation, making it effective even in complex and high-dimensional spaces.

One of Optuna’s key features is its ability to handle multi-objective optimization, enabling users to simultaneously optimize multiple metrics.

The framework is designed to be easily integrated into existing workflows, offering seamless compatibility with popular machine learning libraries such as TensorFlow (which we use here) and PyTorch. Furthermore, Optuna’s intuitive interface facilitates the definition of hyperparameter search spaces and objectives, making it accessible to both novice and experienced practitioners.

5.3.1 Models

We use three models to train on our data, each with their own configurations and objective functions to use in conjunction with Optuna.

In this study, we implemented three different classification models: Decision Tree, XGBoost, and Long Short-Term Memory (LSTM). Each model was designed to capture various patterns in the data and was optimized using different techniques. Below, we describe the objective functions and architectures used for each model.

1. Decision Tree The objective function for the Decision Tree model is defined as follows:

```
def objective_decision_tree(trial, X_train, y_train, X_test, y_test):
    param = {
        'max_depth': trial.suggest_int('max_depth', 3, 20),
        'min_samples_split': trial.suggest_int('min_samples_split',
        2, 20),
        'min_samples_leaf': trial.suggest_int('min_samples_leaf', 1, 20),
        'criterion': trial.suggest_categorical('criterion',
        ['gini', 'entropy']),
    }
    ...
```

Parameter Ranges:

- **max_depth:** This parameter controls the maximum depth of the tree, ranging from 3 to 20. A deeper tree can model more complex relationships but may also lead to overfitting.
- **min_samples_split:** This parameter, ranging from 2 to 20, specifies the minimum number of samples required to split an internal node, impacting the tree’s growth.
- **min_samples_leaf:** This controls the minimum number of samples that must be present in a leaf node, with a range of 1 to 20, preventing the model from creating overly specific rules.
- **criterion:** This parameter defines the function to measure the quality of a split, with options for "gini" or "entropy".

2. XGBoost The objective function for the XGBoost model is outlined as follows:

```
def objective_xgboost(trial, X_train, y_train, X_test, y_test):
    param = {
        'objective': 'multi:softprob',
        'num_class': 3,
        'max_depth': trial.suggest_int('max_depth', 3, 9),
        'learning_rate': trial.suggest_float('learning_rate', 1e-3, 0.3),
        'n_estimators': trial.suggest_int('n_estimators', 50, 500),
        'subsample': trial.suggest_float('subsample', 0.5, 1.0),
        'colsample_bytree':
            trial.suggest_float('colsample_bytree', 0.5, 1.0),
        'gamma': trial.suggest_float('gamma', 0, 5),
        'lambda': trial.suggest_float('lambda', 1e-3, 10.0),
        'alpha': trial.suggest_float('alpha', 1e-3, 10.0),
    }
    ...
```

Parameter Ranges:

- **max_depth**: Controls the maximum depth of the trees, ranging from 3 to 9.
- **learning_rate**: This parameter, ranging from 1×10^{-3} to 0.3, determines the step size at each iteration while moving toward a minimum of the loss function.
- **n_estimators**: This represents the number of trees in the ensemble, ranging from 50 to 500.
- **subsample**: Controls the fraction of samples to be used for fitting the individual base learners, with values between 0.5 and 1.0.
- **colsample_bytree**: Specifies the subsample ratio of columns when constructing each tree, also ranging from 0.5 to 1.0.
- **gamma**: This parameter, ranging from 0 to 5, specifies the minimum loss reduction required to make a further partition on a leaf node.
- **lambda** and **alpha**: These are regularization parameters that control overfitting, with ranges of 1×10^{-3} to 10.0 for both. We also apply early stopping to it by setting a threshold of 5.

3. LSTM Model The LSTM model architecture is constructed to capture temporal dependencies in sequential data. The function `build_lstm_model` is defined as follows:

```
def build_lstm_model(input_shape, num_classes, dropout_rate=0.3):
    model = Sequential()
    model.add(LSTM(50, input_shape=input_shape, return_sequences=False))
    model.add(Dense(50, activation='relu'))
    model.add(Dropout(dropout_rate))
    model.add(Dense(num_classes, activation='softmax'))
    return model
```

Model Architecture:

- **LSTM Layer:** The model starts with an LSTM layer containing 50 units, designed to capture sequential patterns.
- **Dense Layer:** A dense layer with 50 units and a ReLU activation function follows the LSTM layer, allowing for non-linear transformations.
- **Dropout Layer:** A dropout layer with a default rate of 0.3 is included to reduce overfitting.
- **Output Layer:** The final output layer is a dense layer with softmax activation, suitable for multi-class classification tasks (or sigmoid for binary).

Compilation: The model is compiled using the Adam optimizer and the sparse categorical crossentropy loss function, tracking accuracy as a metric.

Training Callbacks: The training process incorporates TensorBoard for real-time monitoring and early stopping to prevent overfitting.

In summary, these models employ various techniques and parameters to optimize performance on the given classification tasks, each catering to different aspects of the data's structure.

5.4 Training

We illustrate a training example by taking an excerpt from the function performing a binary classification task with an Optuna-Tuned XGBoost. We use the same pipeline for the other models as well. For this exercise, the labels are mapped into binary classes. The process is repeated for each subject, where that subject is left out of the training data and is used as a test set.

The code iterates over all subjects, treating each one as the test subject in turn:

```
1 for test_subject in subjects_features.keys():
2     # Exclude test_subject from training data
3     X_train = [features for s, features in subjects_features.items() if s !=
                 test_subject]
```

The `X_train` matrix holds the features from all subjects except the current test subject, ensuring that each subject is left out once during testing.

5.4.1 Label Mapping for Binary Classification

The original labels from the dataset are mapped into two binary classes to simplify the classification task. In this setup, labels '1' are mapped to class '0', and labels '2' and '3' are combined into class '1'. This mapping is applied to both the training and testing labels before fitting the model.

```
1 # Apply label mapping for binary classification
2 y_train = combine_labels(y_train)
3 y_test = combine_labels(y_test)
```

This ensures that the model is trained for binary classification, aligning the labels with the structure of the binary classifier.

5.4.2 Hyperparameter Tuning with Optuna

Hyperparameter tuning is performed using Optuna, which optimizes the XGBoost model's parameters based on the training data for each fold (i.e., each test subject). Optuna performs a search for the best parameters by maximizing the validation accuracy on the left-out test subject.

```
1 study.optimize(lambda trial: objective_xgboost(trial, X_train, y_train, X_test,
        y_test), n_trials=n_trials)
```

The `objective_xgboost` function defines the objective to optimize during training which, like maximizing classification accuracy. This allows the XGBoost model to adapt its parameters dynamically for each subject's specific test case.

5.4.3 Model Training and Evaluation

Once the optimal hyperparameters are found, the XGBoost classifier is trained on the sampled training data, and predictions are made for the left-out test subject. During this process, the training data is sampled using stratified sampling to ensure the class distribution is maintained.

The model's performance is monitored through metrics such as log loss and classification error, which are computed and saved after training:

```
1 best_model.fit(X_train_sampled, y_train_sampled, eval_set=eval_set)
2 y_pred = best_model.predict(X_test)
3 accuracy = accuracy_score(y_test, y_pred)
```

The accuracy for each subject is calculated by comparing the predicted labels (`y_pred`) to the true labels (`y_test`), and these results are logged.

5.4.4 Performance Visualization

The model's performance is visualized by plotting the log loss and classification error over the training epochs for both the training and validation sets. This allows for easy monitoring of overfitting or underfitting during the training process.

```
1 plt.plot(x_axis, results['validation_0']['logloss'], label='Train')
2 plt.plot(x_axis, results['validation_1']['logloss'], label='Validation')
```

5.4.5 Overall Accuracy and Report

After all subjects have been used as test subjects, the overall accuracy across all subjects is calculated. A classification report is generated, which provides precision, recall, and F1 scores for each class:

```
1 overall_accuracy = accuracy_score(all_y_true, all_y_pred)
2 print(classification_report(all_y_true, all_y_pred, target_names=['Class 0', '
        Class 1']))
```

5.4.6 Fusion

In this section, we describe the process of late fusion used to combine predictions from two separate XGBoost models trained on physiological data from wrist and chest sensors. We do the same for the decision tree model as well. The fusion approach aims to enhance the accuracy of the classification by aggregating predictions from both modalities. We focus on the binary prediction model.

Loading Model Predictions The function `load_model_predictions` is responsible for loading the pre-trained XGBoost models and generating predictions on the test data. This function is defined as:

```
1 def load_model_predictions(model_path, X):
2     model = joblib.load(model_path)
3     y_pred = model.predict(X)
4     return y_pred
```

Here, the model is loaded from a specified file path, and the predictions are generated based on the input features X . These predictions are returned and used in the fusion process.

Late Fusion Implementation The `late_fusion` function combines the predictions from wrist and chest models using a simple voting mechanism. The process works as follows:

1. For each subject, the corresponding wrist and chest models are loaded using the `load_model_predictions` function.
2. The feature sets X_{test_wrist} and X_{test_chest} for the wrist and chest models are extracted, along with the true labels y_{test} .
3. The labels are then transformed to ensure binary classification by combining classes 2 and 3 into a single class, as shown:

```
1 y_test = np.where(np.isin(y_test, [1]), 0, 1)
```

4. Predictions are obtained from both models:

```
1 wrist_predictions =
2 load_model_predictions(wrist_model_path, X_test_wrist)
3 chest_predictions =
4 load_model_predictions(chest_model_path, X_test_chest)
```

5. Fusion is performed by adding the predictions and applying a threshold to classify the result:

```
1 fused_predictions = np.where(wrist_predictions + chest_predictions >=
    1, 1, 0)
```

This simple voting mechanism ensures that if either model predicts a positive class (1), the fused prediction will be positive.

6. The fused predictions and true labels are stored for subsequent evaluation.

Evaluation The fused predictions are evaluated using accuracy and a classification report. The results for all subjects are aggregated, and the performance is measured across the dataset:

```
1 overall_accuracy = accuracy_score(all_y_true, all_y_pred_fusion)
2 print(classification_report(all_y_true, all_y_pred_fusion, target_names=['Class
    1', 'Class 2']))
```

Summary In summary, the late fusion technique enables the integration of predictions from two different sensor modalities (wrist and chest). The voting mechanism aggregates the predictions, providing a robust decision framework for the classification task. The fused model’s performance is evaluated using standard metrics like accuracy and the F1-score to assess its effectiveness across different subjects.

6 Results

6.1 Limited Training Data

The first tests running models locally resulted in sub optimal scores. Without any cross-validation techniques, training a Random Forest classifier on a combined set of features from all the subjects, we run into memory allocation errors, and as such, we are unable to use the whole dataset. cutting it short, in order to preserve the temporal nature of the data, we train it using subsets of subjects at a time. For instance, table 10 illustrates the aforementioned random forest model, when we train on a single subject, and use another as a validation set (in this case S1, and S2). We do not continue with this model for further testing:

Table 10: Random Forest Model Run Locally with Limited Training Data

Metric	Precision	Recall	F1-Score	Support
Baseline	0.94	0.82	0.87	826700
Stress	0.58	1.00	0.73	506100
Amusement	0.00	0.00	0.00	260400
Accuracy	0.74			
Macro Avg	0.51	0.61	0.54	1593200
Weighted Avg	0.67	0.74	0.69	23,206,404

As we see, there is overfitting occurring. This could be due to a few reasons. The first being the temporal nature of physiological data in general, because of which the model might be memorizing instead of learning. Switching to a system with a 20-core processor and 128 gigabytes of memory, it became possible to test out LOSOCV with the entire dataset.

6.2 Baseline

We have the results of our decision tree baseline using the LOSOCV (which from here on out is implied for every model) is provided in tables 11 and 12. The results

are akin to random guessing, with the model falling well short of the 50 percent mark in accuracy.

Table 11: Classification Report for baseline on wrist

Metric	Precision	Recall	F1-Score	Support
Baseline	0.54	0.56	0.55	12,327,702
Stress	0.35	0.30	0.32	6,976,201
Amusement	0.14	0.16	0.15	3,902,501
Accuracy	0.41			
Macro Avg	0.34	0.34	0.34	23,206,404
Weighted Avg	0.41	0.41	0.41	23,206,404

Table 12: Classification Report for baseline model on chest

Metric	Precision	Recall	F1-Score	Support
Baseline	0.63	0.57	0.60	12,327,702
Stress	0.39	0.51	0.44	6,976,201
Amusement	0.22	0.17	0.19	3,902,501
Accuracy	0.48			
Macro Avg	0.41	0.42	0.41	23,206,404
Weighted Avg	0.49	0.48	0.48	23,206,404

6.3 Hyperparameter Optimization

When using Optuna, with the configurations mentioned in section 5.3.1, the results vastly improve. We are no longer guessing, and we see from table 13 that for the wrist modality, there is a jump of 12 percent.

Table 13: XGBoost with Optuna on chest

Metric	Precision	Recall	F1-Score	Support
Baseline	0.70	0.71	0.70	12,327,702
Stress	0.52	0.65	0.58	6,976,201
Amusement	0.26	0.13	0.17	3,902,501
Accuracy	0.59			
Macro Avg	0.49	0.50	0.48	23,206,404
Weighted Avg	0.57	0.59	0.58	23,206,404

We test out the XGBoost model, outlined in section 5.3.1. We also apply early stopping to it in the hopes of reducing training times. This is discussed further in section 7.2. While the wrist accuracies were in line with the decision tree’s performance, the chest data saw a rise up to 59 percent.

6.4 Dataset Reduction

In a bid to reduce training times, we attempt to reduce the size of the training data, down to a 25 percent of the original. Surprisingly, while this reduced training time by a vast margin, the predictive ability of the models were not proportionally affected, offering competitive results as seen in table 14. This could be due to the cross-validation approach. LOSO works quite well with limited data. In this case, we go over the same dataset 15 times, during which the model might be training on the same data multiple time, leading to redundancies. Limiting the training data could allow the model to learn from the fullest data possible while still preserving generalizability. This is especially useful when data collection is expensive or limited, as in the case of multi-modal physiological recordings.

Table 14: XGBoost tuned, with a reduced dataset on chest

Metric	Precision	Recall	F1-Score	Support
Baseline	0.66	0.73	0.70	12,327,702
Stress	0.54	0.62	0.58	6,976,201
Amusement	0.09	0.04	0.06	3,902,501
Accuracy	0.58			
Macro Avg	0.43	0.46	0.44	23,206,404
Weighted Avg	0.53	0.58	0.55	23,206,404

6.4.1 Long Short Term Memory (LSTM)

We also apply LSTM here. LSTMs are well-suited for time-series data because they capture long-term dependencies in sequences, which is essential for physiological data that evolves over time. They can effectively handle fluctuations in heart rate, respiration, and other physiological signals, learning temporal patterns associated with stress or amusement.

We keep the model relatively simple. LSTMs have a high computational cost due to their sequential nature, where each time step depends on the previous one. This makes them slow to train, especially with long sequences or large datasets like WESAD. This can lead to long training times, increased memory usage, and the need for high-performing hardware, especially when dealing with multiple sensor modalities. We can observe in figure 4, the effects of early stopping, where for both subjects we terminate before the 10 epochs are reached.

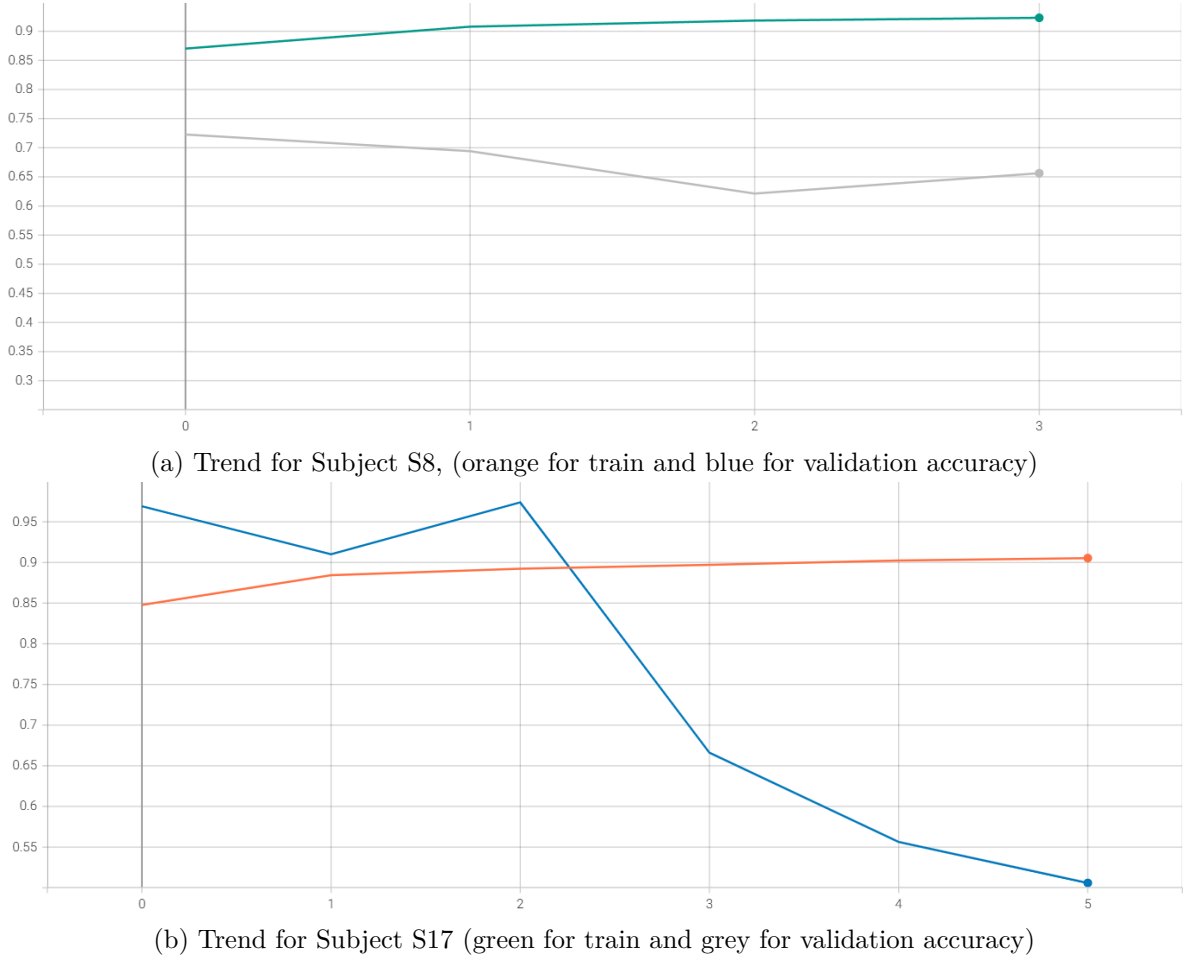


Figure 3: Visualization of LSTM accuracies over epochs on a binary classification task with chest data

6.5 Binary Classification

A trend noticed in the classification reports of the models tested previously is the particularly low scores in predicting the 'amusement' state. Every model tested so far almost completely fail to predict when a subject is amused, and always incorrectly categorized them as 'stressed'. This was noted by Schmidt et al. as well, for which they decided to merge the stress and amusement labels as a single stressed or 'non-baseline' label [8]. The results of this binary classification problem now reveal a much clearer distinction between the states. We combined our previously applied optimization techniques on this task (i.e. Optuna with a dataset reduction).

The results now are overwhelmingly in favour of the decision tree model, especially given the time required for training in comparison to the others as we will see. The best result, which is on the chest data, however, comes from the LSTM model, which does seem to be the most suitable application for this task on paper with an accuracy of 75 percent, however, it can only outperform the decision tree model by a small margin (2 percent).

Table 15: Classification Report (Binary) for Tuned Decision Tree With Reduced Dataset on Chest

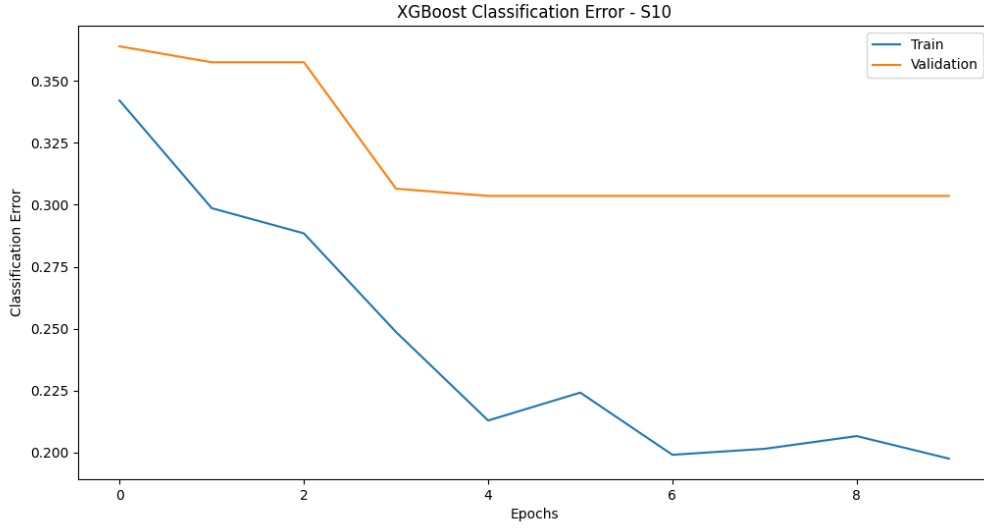
Metric	Precision	Recall	F1-Score	Support
Baseline	0.76	0.73	0.74	12,327,702
Stress	0.71	0.74	0.72	10,878,702
Accuracy	0.73			
Macro Avg	0.73	0.73	0.73	23,206,404
Weighted Avg	0.73	0.73	0.73	23,206,404

Table 16: Classification Report (Binary) for Tuned Decision Tree With Reduced Dataset on Wrist

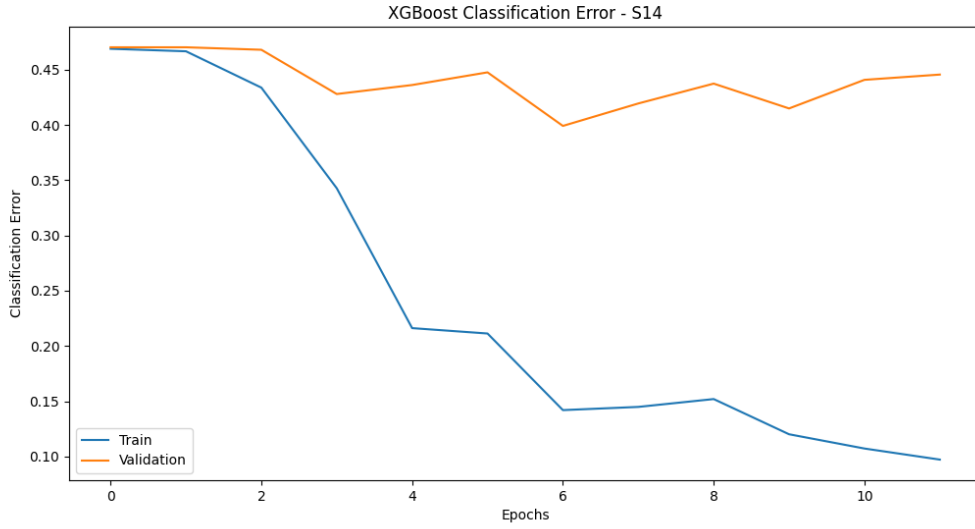
Metric	Precision	Recall	F1-Score	Support
Baseline	0.70	0.73	0.72	12,327,702
Stress	0.68	0.65	0.67	10,878,702
Accuracy	0.69			
Macro Avg	0.69	0.69	0.69	23,206,404
Weighted Avg	0.69	0.69	0.69	23,206,404

An overview of the decision tree model can be found in table 15. Additionally, we see from the classification matrix of the decision tree model, that it can much more clearly classify a subject into a baseline or stressed state, as evidence by the precision and recall scores.

We also present some multiclassification error curves for subjects S10 and S14, using the XGBoost model. It is calculated as $No.of\ wrong\ cases / Total\ cases$. We see how with early stopping, S10 terminates at around the 9th epoch.



(a) Classification error for Subject S10



(b) classification error for Subject S14

Figure 4: Visualization of XGBoost classification errors over epochs on a binary classification task with wrist data

6.6 Fusion

The results for late fusion on our most optimal decision tree model can be found below:

Table 17: Binary Classification Report for Late Fusion on Decision Tree

Metric	Precision	Recall	F1-Score	Support
Baseline	0.80	0.55	0.65	12,327,702
Stress	0.62	0.85	0.72	10,878,702
Accuracy	0.69			
Macro Avg	0.71	0.70	0.69	23,206,404
Weighted Avg	0.72	0.69	0.68	23,206,404

6.7 Observed trends

Analyzing the accuracy plots for chest and wrist models across subjects highlights the variability and overall performance trends of both Decision Tree and XGBoost models, as seen in figure 5. Subjects like S2, S3, and S5, XGBoost’s performance remains relatively high, whereas Decision Trees show more fluctuation and lower accuracy, particularly in chest models for these subjects. This disparity can be attributed to XGBoost’s ability to handle complex patterns in the data, while Decision Trees, especially without tuning, are more prone to underfitting, hence showing reduced performance on test instances.

Interestingly, for some subjects such as S4 and S11, the wrist models appear to consistently achieve higher accuracy than the chest models. This pattern could suggest that for these specific subjects, the physiological data captured by wrist sensors are more predictive or less noisy than chest sensor data. However, variability across subjects remains, especially in the Decision Tree wrist model, which sometimes achieves comparable accuracy to chest models but shows irregular performance across subjects like S14 and S15.

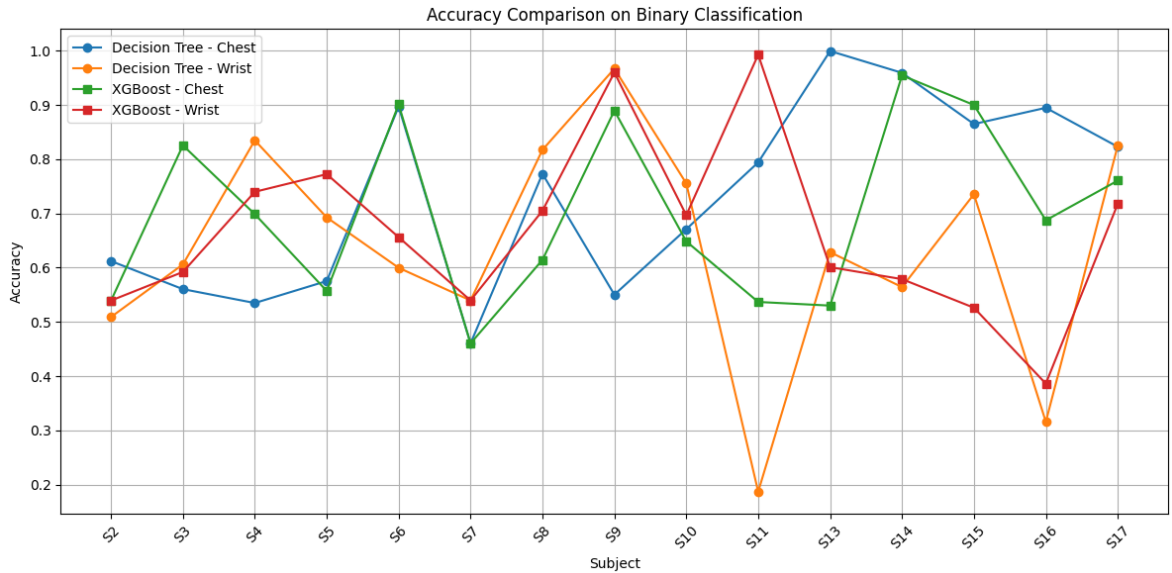


Figure 5: Accuracy Trend for all Subjects with Optuna applied

6.8 Comparison with Reference Paper

Schmidt et al. using the same cross-validation technique reached 80 percent on the chest data with AdaBoost for the multi class problem, and 93 percent with LDA on the chest data with the binary classification problem. Although we do not reach the same overall accuracy, we do hit these numbers on certain subjects with various models as seen in figure 5.

Model	AdaBoost	LDA	Random Forest	Decision Tree	LSTM	Decision Tree
Source	Paper	Paper	Paper	Own	Own	Own
Acc. (%)	0.80	0.93	0.76	0.72	0.75	0.67
Problem	Multi	Binary	Multi	Binary	Binary	Multi
Modality	Chest	Chest	Wrist	Chest	Chest	Chest

Figure 6: Comparison of results with Schmidt et al.

7 Discussion

7.1 Data Availablitiy

Owing to the short sample size of the pilot study dataset, we were unable to perform any predictive modelling on it. Out of the 3 subjects, all of them were non-depressive. Instead, we focused our efforts on the WESAD dataset. Despite the labels not being the same, it was the dataset closest in structure to the pilot study, which meant that the preprocessing and modelling pipeline could (for the most part) be easily carried over to fit to the pilot study, in the event more data is added to it.

7.2 Overfitting

There were some initial problems faced working with the WESAD dataset that necessitated corrective measures. The main one being the nature of the data and how predictive modelling behaved on it. In every case, using a simple 80-20 train-test split, led to overfitting. Expanding this to include all 15 subjects gave the exact same results.

Trying out multiple different models made no difference. Keeping the pre-processing pipeline the same, every model resulted in roughly a 0.95 to 1.00 in accuracy for the multiclass classification problem.

Another method to mitigate this was to attempt shuffling the data. This was once again tried on a per-subject basis and also on all 15 subjects at once with the same results, although, the temporal nature of the data was lost, making it an unsustainable way forward.

7.3 Runtimes

The computational complexity of LOSOCV is considerably higher than that of k-fold cross-validation, as the model must be trained N times, which can be particularly taxing for large datasets and complex models, however in the interest of replicating the workflow showcased by Schmidt et al., and also memory constraints on local machines, we continue with LOSOCV.

This was reflected in the results, and every model besides base decision tree without any hyperparameter optimization took well over ten hours to run per modality, i.e. either one of wrist or chest. When reducing dataset to twenty percent, there were reductions in train times. However, they were still long, as evidenced by the LSTM model on binary classification. Runs without any down-sampling applied often took over twelve hours to run per modality. We can observe this in figure 7. LSTMs thus, were not sustainable to keep testing.

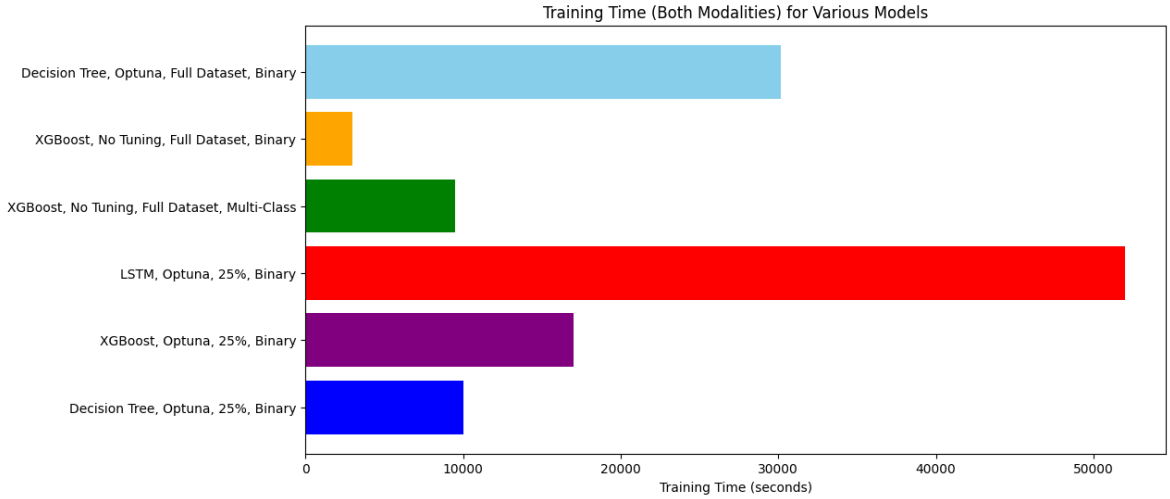


Figure 7: Training time of various models

7.4 Memory Requirements

The nature of the task made it impossible to run any model on local machines. On a personal system, with 8 gigabytes of DDR4 memory at 2666MHz and a quad-core processor, the program was unable to allocate the memory required to train a decision tree on more than two subjects at a time, far off from the required 14 (and 1 left out as a test set). Thus, until access to the remote server was granted, we were unable to run any models on the complete data.

7.5 Differentiating Between Stress and Amusement

As remarked by Schmidt et al., models were indeed found to struggle to distinguish between the classes 2 and 3 (states Stress and Amusement), with exceedingly poor precision for those classes, as they were quite often misclassified as each other. To recapitulate, in line with their approach, we also tried out a binary classification by combining stress and amusement into a single class

8 Outlook

8.1 Conclusions

In this work, with close reference to the dataset and methodology followed by Schmidt et al., we perform an analysis on the WESAD dataset. We apply various models to it, along with relevant cross-validation techniques and hyperparameter optimization.

The binary classification problem was a result of a high misclassification rate in the 'amusement' class, for which we merge two labels into one general 'stressed' label.

Futhermore, We find that we can further optimize performance by reducing our training data to 25 percent, without a substantial hit to the accuracy. The best score we were able to reach was with the LSTM on the chest data on a binary classification problem with an accuracy of 0.75. In comparison, for the same task, Schmidt et al. managed an accuracy of 0.95 [8]

8.2 Future work

While the results achieved in this study demonstrate promising insights, several avenues for future work remain to enhance both the effectiveness and generalizability of the proposed methods.

Investigating advanced fusion techniques beyond late fusion, such as intermediate or hybrid fusion approaches, may offer a deeper understanding of the complementary information within each modality. Such methods could allow models to dynamically adapt to different subjects or scenarios, potentially improving classification accuracy across a wider range of contexts.

In terms of model optimization, exploring other machine learning techniques, such as ensemble methods or attention-based neural networks, could further improve classification performance and interpretability. For the decision tree-based models, applying advanced hyperparameter optimization techniques may yield a more optimized trade-off between accuracy and computational efficiency, especially for real-time applications.

Finally, expanding the current framework to include a larger and more diverse dataset would enhance model robustness and generalizability across different populations. Future work could also involve transferring this approach to other applications within healthcare and behavioral studies, where detecting and interpreting physiological responses is crucial. By pursuing these directions, it may be possible to achieve a more comprehensive and flexible framework with improved utility in real-world scenarios.

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